Ch-3 (Demand Theory Application)

Q-16: (a) Distinguish between CV and EV.

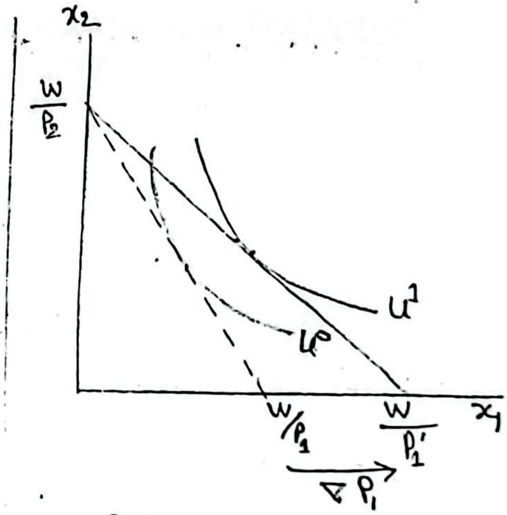
(b) How do you measure the welfare effects of a price change

Ans: ~~(b)~~ Measure the welfare effects of a price change – শুধু এই টপিক চাইলে শুরু থেকে লিখব

(a) Distinguishing between CV and EV

① **Compensating Variation (CV):**

This welfare measures how much money would a consumer be willing to give up after a reduction in prices to be just as well off as before the price decrease. For this reason we refer this measure as the "compensating variation", since it represents the change in income that would exactly compensate the consumer for the impact that the price change has on his welfare. Importantly, this measure maintains the initial utility level, u (before any price change), ~~~~.



The fig. represents a decrease in the price of x_1, from P_1 to P_1' yielding an increase in the utility level the consumer reaches after solving his utility maximization problem (UMP), from u^0 to u^1.

* বেশি ~~compen~~ CV এর fig. না* । এই figure থেকে CV আসবে হবে। পরের page-৩ CV এর fig আছে। এই Fig. price decrease and utility র relation দেখায়

② Equivalent variation (EV): It responds to the question: How much money would a consumer need before a reduction in prices to be as well off as after the price decrease? For this reason we refer to this measure as the 'equivalent variation,' since it reflects the change in income that would be equivalent to the impact that the price change has on his welfare.

In this case, we focus on the final utility level that the individual can reach after price change, u^1.

(b) Measure the welfare effects of a price change:

Two approaches:

① CV and EV Using expenditure function

② CV and EV Using Hicksian demand.

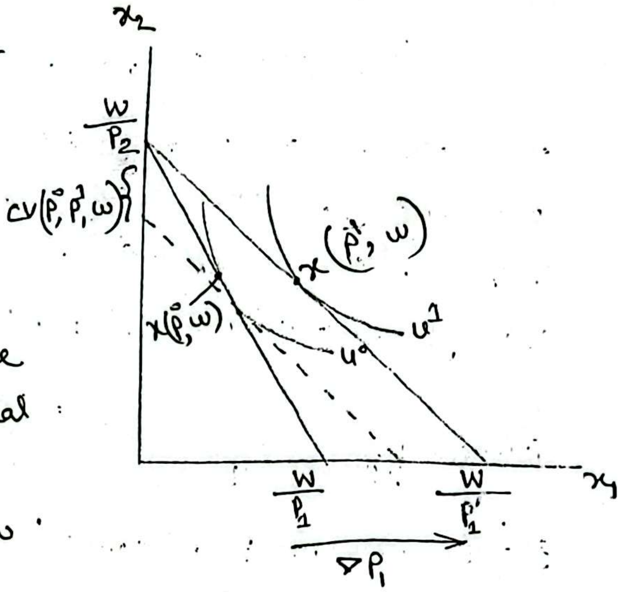
① CV using Expenditure Function:

$CV(p^0, p^1, w)$ using $e(p, u)$, expenditure function we get,

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0)$$

The amount of money the consumer is willing to give up after the price decrease (after price level is p^1 and her utility level has improved to u^1) to be just as well off as before the price decrease (reaching utility level u^0)

The figure represents the compensating variation using the expenditure function.



Here is a basic procedure on how to find the CV from a price decrease. Before the price change, the consumer selects optimal bundle $x(p^0, w)$ reaching utility level u^0 under budget set $B_{p^0, w}$.

After p^1 decreases, the consumer choses optimal bundle $x(p^1, w)$ under budget set $B_{p^1, w}$. We then adjust the consumer's wealth to make him just as well off as before the price change. We represent this adjustment by a parallel shift in budget line $B_{p^1, w}$, to the dashed budget line, where the individual can reach his initial utility level u^0 at the final price ratio. At this point his expenditure is $e(p^1, u^0)$. Hence the difference in expenditure $e(p^1, u^1)$ at $B_{p^1, w}$ and $e(p^1, u^0)$ at the dashed line — represents the compensating variation resulting from the price change.

In short (সংক্ষেপে),

① When $B_{p^0, w}$, $x(p^0, w)$

② ∇p_1 and $x(p^1, w)$ under $B(p^1, w)$

③ Adjust final wealth (after the price change to make the consumer as well off as before the price change.

④ Difference in expenditure

$$CV(p^0, p^1, w) = \underbrace{e(p^1, u^1)}_{\text{at } B_{p^1, w}} - \underbrace{e(p^1, u^0)}_{\text{dashed}}$$

This is Hicksian wealth compensation.

EV using Expenditure Function :

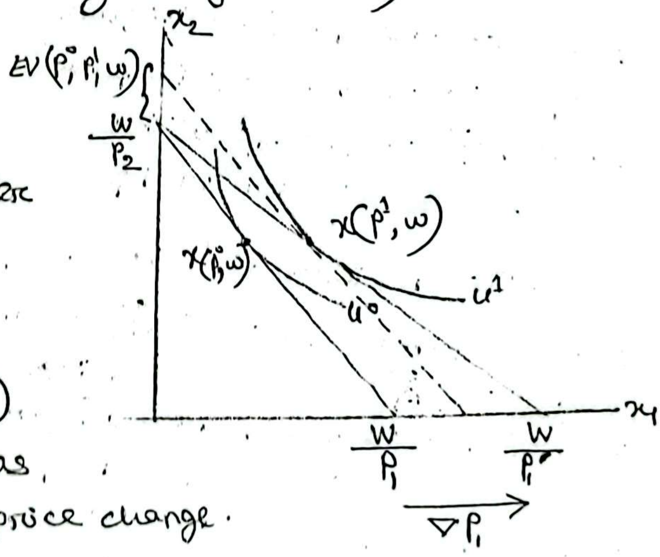
$EV(p^0, p^1, w)$, using expenditure function $e(p, u)$:

$$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0).$$

The amount of money the consumer needs to receive before the price decrease (at the initial price level p^0 when her utility level is still u^0) to be just as well off as after the price decrease (reaching utility level u^1).

① when $B(p^0, w)$, $x(p^0, w)$

② ∇p_1 and $x(p^1, w)$ under $B(p^1, w)$.



③ Adjust initial wealth (before the price change) to make the consumer as well off as after the price change.

④ Difference in expenditure:

$$EV(p^0, p^1, w) = \underbrace{e(p^0, u^1)}_{\text{dashed line}} - \underbrace{e(p^0, u^0)}_{\text{at } B(p^0, w)}$$

②

CV using Hicksian Demand :

From the previous definitions we know that if $p_1^1 < p_1^0$ and $p_k^1 = p_k^0$ for all $k \neq 1$, then

$\boxed{e(p^1, u^1) = w}$

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0)$$
$$= w - e(p^1, u^0)$$
$$= e(p^0, u^0) - e(p^1, u^0)$$
$$= \int_{p_1^1}^{p_1^0} \frac{\delta e(p_1, \bar{p}_{-1}, u^0)}{\delta p_1} \cdot dp_1$$
$$= \int_{p_1^1}^{p_1^0} h_1(p_1; \bar{p}_{-1}, u^0)\, dp_1$$

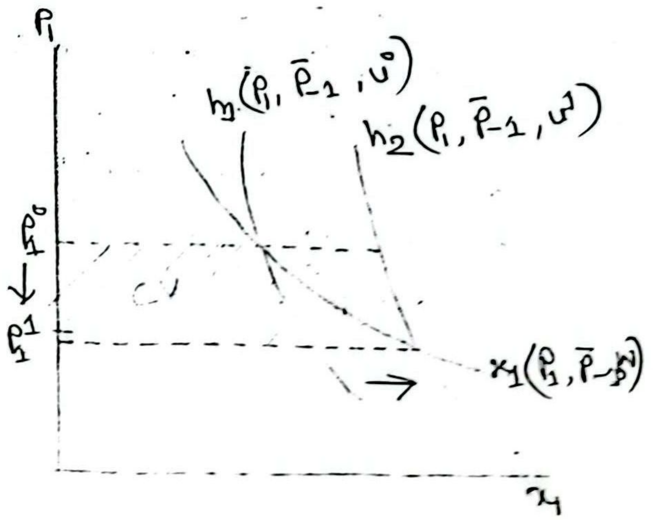
Since $e(p^1, u^1) = e(p^0, u^0) = w$ and $\frac{\delta e(p, u)}{\delta p_1} = h_1(p, u)$.

In addition, let $\bar{p}_{-1}$ denote the price vector of all other goods $k \neq 1$ such that $\bar{p}_{-1} = (p_2, p_3, \ldots, p_L)$, which are unaffected by the price change.

The case is:
- Normal good
- Price decrease

Graphically, CV is represented by the area to the left of the Hicksian demand curve for good 1 associated with utility level u^0 and lying between prices p_1^1 and p_1^0.

The welfare gain is represented by the shaded region.



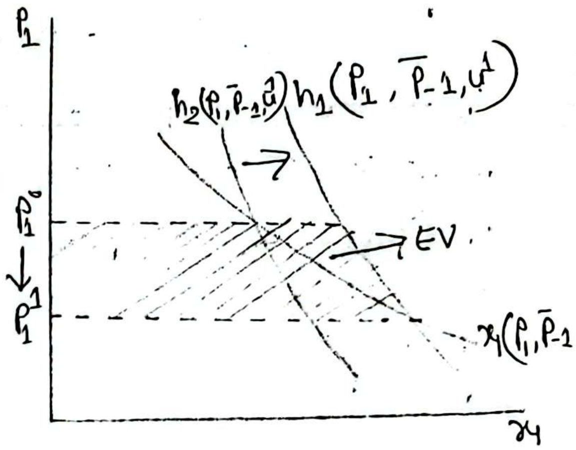
EV using Hicksian Demand :

From the previous definitions we know that, if $p_1^1 < p_1^0$ and $p_k^1 = p_k^0$ for all $k \neq 1$, then

$$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0)$$
$$= e(p^0, u^1) - w$$
$$= e(p^0, u^1) - e(p^1, u^1)$$
$$= \int_{p_1^1}^{p_1^0} \frac{\delta e(p_1, \bar{p}_{-1}, u^1)}{\delta p_1} \, dp_1$$
$$= \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^1) \, dp_1 .$$

The case is:
- Normal good
- Price decrease

Graphically, EV is represented by the area to the left of the Hicksian demand curve for good 1 associated with utility level u^1 and lying between prices p_1^1 and p_1^0.



The welfare gain is represented by the shaded region.

Normally price increase দেখায় না।

What about a price increase?

The Hicksian demand associated with initial utility level u^0 (before the price increase or before the introduction of a tax) experiences an inward shift when the price increases, or consumer's utility level is now u^1, where $u^0 > u^1$. Hence,

$$h_1(P_1, \bar{P}_{-1}, u^0) > h_1(P_1, \bar{P}_{-1}, u^1).$$

The definitions of CV and EV would now be:

CV: the amount of money that a consumer would need after a price increase to be as well off as before the price increase.

EV: the amount of money that a consumer would be willing to give up before a price increase to be as well off as after the price increase.

Q-17: Graphically show that for a decrease in the price of a commodity (which both goods are normal).

EV > CS > CV

for price increase,

EV < CS < CV .

Ans:

Walrasian demand is easier to observe, so we could use the variation in consumer's surplus as an approximation of welfare changes. A decrease in P_1 would imply an increase in CS. This measurement coinsides with that provided by the CV and EV, when income effects are zero:

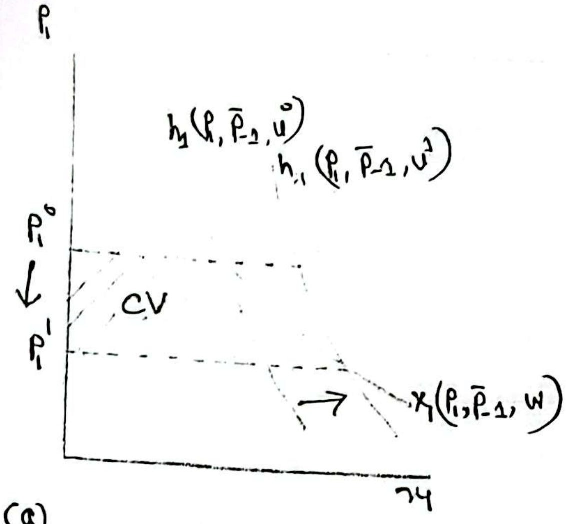
Recall that the Walrasian demand measures both income and substitution effects resulting from a price change, while,

The Hicksian demand measures only substitution effects from such a price change.

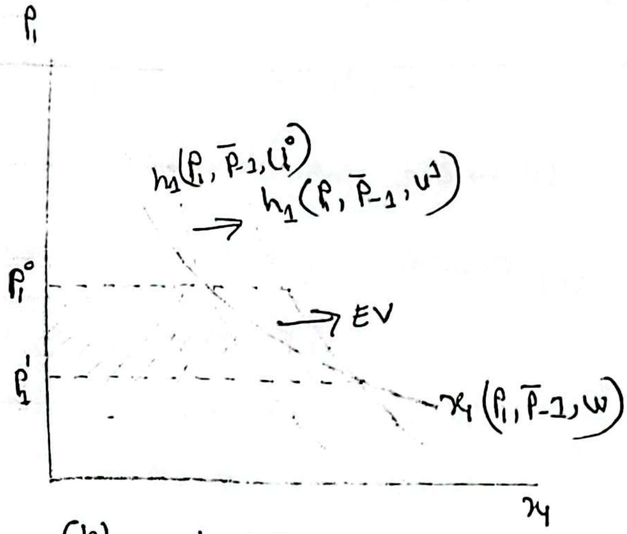
Hence, there will be a difference between CV and CS, and between EV and CS.

The figure illustrates the difference between these welfare measures for the case in which good 1 is a "normal" good, namely income effects are positive. Hence, the Walrasian demand is more sensitive to price changes than the Hicksian demand, implying that CV < CS whereas EV > CS, meaning EV > CS > CV .

• Normal goods



(a) CV < CS



(b) CS < EV

For normal goods:

Price decrease : CV < CS < EV.

Price increase : CV > CS > EV.

For Inferior good

Price decrease: EV < CS < CV

" increase: EV > CS > CV.

Consumer Surplus, $CS = \int_{p_1^1}^{p_1^0} x(p_1, \bar{p}_{-1}, m)\, dp_1$

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Ch-3

Q-22: Consider a consumer with the utility function

$$u(x_1, x_2) = x_1^{1/4} x_2^{1/2}$$

(i) Derive the Walrasian and compensated demand function.

(ii) Consider $\{p^0, m^0\} = \{4, 4, 300\}$

$\{p^1, m^0\} = \{2, 4, 300\}$

Compute the indirect utilities and costs.

(iii) Calculate EV, CV (& CS) and examine their relationship and comment on the nature of the commodities.

Solve for Walrasian Demand

Solution: Given that,

Utility function, $u(x_1, x_2) = x_1^{1/4} x_2^{1/2}$

Budget constraint, $p_1 x_1 + p_2 x_2 = m$.

utility max
max. $u(x)$
s.t. $p.x \leq m_{wealth}$

To maximize utility subject to a budget constraint, we set up a Lagrangian function,

$$L = x_1^{1/4} x_2^{1/2} + \lambda \left[m - p_1 x_1 - p_2 x_2\right]$$

FONC requires:

$$\frac{\delta L}{\delta x_1} = \frac{1}{4} x_1^{\frac{1}{4}-1} x_2^{\frac{1}{2}} - \lambda p_1 = 0$$

$$\Rightarrow \frac{1}{4} x_1^{\frac{-3}{4}} x_2^{\frac{1}{2}} = \lambda p_1 \quad \text{------ (i)}$$

$$\frac{\delta L}{\delta x_2} = \frac{1}{2} x_1^{\frac{1}{4}} x_2^{\frac{1}{2}-1} - \lambda p_2 = 0$$

$$\Rightarrow \frac{1}{2} x_1^{\frac{1}{4}} x_2^{\frac{-1}{2}} = \lambda p_2 \quad \text{------ (ii)}$$

$$\frac{\delta L}{\delta \lambda} = m - P_1 x_1 - P_2 x_2 = 0 \; \ldots\ldots \; (iii)$$

Divide Eq (i) by (ii),

$$\frac{\frac{1}{4} x_1^{-\frac{3}{4}} x_2^{\frac{1}{2}}}{\frac{1}{2} x_1^{\frac{1}{4}} x_2^{-\frac{1}{2}}} = \frac{\lambda P_1}{\lambda P_2}$$

$$\Rightarrow \frac{1}{4} \times \frac{2}{1} \times x_1^{-\frac{3}{4} - \frac{1}{4}} \cdot x_2^{\frac{1}{2} + \frac{1}{2}} = \frac{P_1}{P_2}$$

$$\Rightarrow \frac{1}{2} x_1^{\frac{-3-1}{4}} \cdot x_2^{\frac{1+1}{2}} = \frac{P_1}{P_2}$$

$$\Rightarrow \frac{1}{2} x_1^{-1} \cdot x_2^{1} = \frac{P_1}{P_2}$$

$$\Rightarrow \frac{x_2}{2x_1} = \frac{P_1}{P_2}$$

$$\Rightarrow x_2 = \frac{2x_1 P_1}{P_2}$$

Substituting this value into Eq. (iii),

$$P_1 x_1 + P_2 x_2 = m$$

$$\Rightarrow P_1 x_1 + \cancel{P_2} \frac{2x_1 P_1}{\cancel{P_2}} = m$$

$$\Rightarrow 3 P_1 x_1 = m$$

$$\Rightarrow x_1 = \frac{m}{3P_1}$$

Similarly, $x_2 = \dfrac{2x_1 P_1}{P_2}$

$$= \frac{2 \times \frac{m}{3P_1} P_1}{P_2}$$

$$= \frac{2m}{3} \times \frac{1}{P_2}$$

$$= \frac{2m}{3P_2}$$

Walrasian demand for goods, $x_1 = \dfrac{m}{3P_1}$ and $x_2 = \dfrac{2m}{3P_2}$.

Solve for compensated demand:

The cost function for this consumer is obtained by minimizing cost subject to a utility constraint.

Expenditure min.
min. $P.x$
s.t. $u(x) \geq u$

Lagrangian function:

$$L = P_1 x_1 + P_2 x_2 + \lambda \left[u - x_1^{\frac{1}{4}} x_2^{\frac{1}{2}} \right]$$

FONC requires,

$$\frac{\delta L}{\delta x_1} = P_1 - \frac{1}{4} x_1^{\frac{1}{4}-1} \lambda x_2^{\frac{1}{2}} = 0$$

$$\Rightarrow P_1 = \frac{1}{4} x_1^{\frac{-3}{4}} \lambda x_2^{\frac{1}{2}} \quad \text{------ (i)}$$

$$\frac{\delta L}{\delta x_2} = P_2 - \lambda x_1^{\frac{1}{4}} \frac{1}{2} \cdot x_2^{\frac{1}{2}-1} = 0$$

$$\Rightarrow P_2 = \frac{1}{2}\lambda x_1^{\frac{1}{4}} x_2^{\frac{-1}{2}} \quad \cdots\cdots (ii)$$

$$\frac{\delta L}{\delta \lambda} = u - x_1^{\frac{1}{4}} x^{\frac{1}{2}} = 0 \quad \cdots\cdots (iii)$$

Divide Eq. (i) by (ii),

$$\frac{P_1}{P_2} = \frac{\frac{1}{4}\lambda x_1^{-\frac{3}{4}} x_2^{\frac{1}{2}}}{\frac{1}{2}\lambda x_1^{\frac{1}{4}} x_2^{-\frac{1}{2}}}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{1}{4} \times \frac{2}{1} \cdot x_1^{-\frac{3}{4}-\frac{1}{4}} \cdot x_2^{\frac{1}{2}+\frac{1}{2}}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{1}{2} \cdot \frac{x_2}{x_1}$$

$$\Rightarrow x_2 = \frac{2x_1 P_1}{P_2}$$

Substituting this value into Eq. (iii)

$$x_1^{\frac{1}{4}} x_2^{\frac{1}{2}} = u$$

$$\Rightarrow x_1^{\frac{1}{4}} \left(\frac{2x_1 P_1}{P_2}\right)^{\frac{1}{2}} = u$$

$$\Rightarrow 2^{\frac{1}{2}} \cdot \left(\frac{P_1}{P_2}\right)^{\frac{1}{2}} x_1^{\frac{1}{4}+\frac{1}{2}} = u$$

$$\Rightarrow 2^{\frac{1}{2}} \left(\frac{P_1}{P_2}\right)^{\frac{1}{2}} x_1^{\frac{3}{4}} = u \, .$$

$$\Rightarrow x_1^{\frac{3}{4}} = u\, 2^{-\frac{1}{2}} \left(\frac{p_1}{p_2}\right)^{-\frac{1}{2}}$$

$$\Rightarrow x_1^{\frac{3}{4}} = u\, 2^{-\frac{1}{2}} \left(\frac{p_2}{p_1}\right)^{\frac{1}{2}}$$

$$\Rightarrow x_1 = u^{\frac{4}{3}}\, 2^{-\frac{1}{2}\times\frac{4}{3}} \left(\frac{p_2}{p_1}\right)^{\frac{1}{2}\times\frac{4}{3}}$$

$$\Rightarrow x_1 = u^{\frac{4}{3}} \cdot 2^{-\frac{2}{3}} \cdot \left(\frac{p_2}{p_1}\right)^{\frac{2}{3}}$$

$$\Rightarrow x_1 = u^{\frac{4}{3}} \cdot \left(\frac{1}{2}\right)^{\frac{2}{3}} \cdot \left(\frac{p_2}{p_1}\right)^{\frac{2}{3}}$$

Similarly for,

$$x_2 = u^{\frac{4}{3}} \cdot 2^{\frac{1}{3}} \cdot \left(\frac{p_1}{p_2}\right)^{\frac{1}{3}}$$

$$x_2 = \frac{2 x_1 p_1}{p_2}$$

$$= \frac{2 \cdot u^{\frac{4}{3}} \left(\frac{1}{2}\right)^{\frac{2}{3}} \cdot \left(\frac{p_2}{p_1}\right)^{\frac{2}{3}} \cdot p_1}{p_2}$$

$$= 2 \cdot u^{\frac{4}{3}} \cdot 2^{-\frac{2}{3}} \cdot p_2^{\frac{2}{3}-1} \cdot p_1^{1-\frac{2}{3}}$$

$$= 2^{1-\frac{2}{3}} \cdot u^{\frac{4}{3}} \cdot p_2^{\frac{2-3}{3}} \cdot p_1^{\frac{3-2}{3}}$$

$$= 2^{\frac{4}{3}} \cdot u^{\frac{4}{3}} \cdot p_2^{-\frac{1}{3}} \cdot p_1^{\frac{1}{3}}$$

$$= u^{\frac{4}{3}}\, 2^{\frac{1}{3}} \cdot \left(\frac{p_1}{p_2}\right)^{\frac{1}{3}}$$

So, compensated / Hicksian demand for goods, $x_1 = u^{\frac{4}{3}} \left(\frac{1}{2}\right)^{\frac{2}{3}} \left(\frac{p_2}{p_1}\right)^{\frac{2}{3}}$

and, $x_2 = u^{\frac{4}{3}} \cdot 2^{1/3} \left(\frac{p_1}{p_2}\right)^{\frac{1}{3}}$

(ii) Given that,

$$\{p^0, m^0\} = \{4, 4, 300\}$$

$$\{p^1, m^0\} = \{2, 4, 300\}$$

Indirect Utility

we get from (i),

Walrasian demand function for two goods.

$$x_1 = \frac{m}{3p_1}, \quad x_2 = \frac{2m}{3p_2}$$

Indirect utility function:

$$\psi(m, p_1, p_2) = x_1^{\frac{1}{4}} \cdot x_2^{\frac{1}{2}}$$

$$= \left(\frac{m}{3p_1}\right)^{\frac{1}{4}} \left(\frac{2m}{3p_2}\right)^{\frac{1}{2}}$$

$$= 2^{\frac{1}{2}} \cdot 3^{-\frac{1}{4}-\frac{1}{2}} \cdot \left(\frac{m}{p_1}\right)^{\frac{1}{4}} \left(\frac{m}{p_2}\right)^{\frac{1}{2}}$$

$$= \sqrt{2} \cdot 3^{\frac{-3}{4}} \left(\frac{m}{p_1}\right)^{\frac{1}{4}} \left(\frac{m}{p_2}\right)^{\frac{1}{2}}$$

$$= \frac{\sqrt{2}}{3^{3/4}} \left(\frac{m}{p_1}\right)^{\frac{1}{4}} \left(\frac{m}{p_2}\right)^{\frac{1}{2}}$$

$$u_0 = \psi_0 \{ p^0, m^0 \} = \frac{\sqrt{2}}{3^{3/4}} \cdot \left(\frac{m}{p_1}\right)^{\frac{1}{4}} \cdot \left(\frac{m}{p_2}\right)^{\frac{1}{2}}$$

$$= \frac{\sqrt{2}}{3^{3/4}} \left(\frac{300}{4}\right)^{\frac{1}{4}} \cdot \left(\frac{300}{4}\right)^{\frac{1}{2}}$$

$$= \frac{\sqrt{2}}{3^{3/4}} \cdot 75^{\frac{1}{4}} \cdot 75^{\frac{1}{2}}$$

$$= \frac{\sqrt{2}}{3^{\frac{3}{4}}} \cdot 75^{\frac{3}{4}}$$

$$= 15.811$$

$$u_1 = \psi_1 \{ p^1, m^1 \} = \frac{\sqrt{2}}{3^{3/4}} \cdot \left(\frac{300}{2}\right)^{\frac{1}{4}} \cdot \left(\frac{300}{4}\right)^{\frac{1}{2}}$$

$$= 18.803$$

Now compute various measures of cost.

Costs :

From compensated demand, we get,

$$x_1 = u^{\frac{4}{3}} \left(\frac{1}{2}\right)^{\frac{2}{3}} \left(\frac{P_2}{P_1}\right)^{\frac{2}{3}}$$ and

$$x_2 = u^{4/3}\, 2^{\frac{1}{3}} \left(\frac{P_1}{P_2}\right)^{\frac{1}{3}}.$$

Cost function: Substitute x_1 and x_2 in the cost f^n,

$$c = P_1 x_1 + P_2 x_2$$

$$= P_1\, u^{\frac{4}{3}} \left(\frac{1}{2}\right)^{\frac{2}{3}} \left(\frac{P_2}{P_1}\right)^{\frac{2}{3}} + P_2\, u^{4/3} . 2^{1/3} \left(\frac{P_1}{P_2}\right)^{\frac{1}{3}}$$

$$= u^{\frac{4}{3}} \left[\left(\frac{1}{2}\right)^{\frac{2}{3}} P_1^{1-\frac{2}{3}} . P_2^{\frac{2}{3}} + 2^{\frac{1}{3}} P_1^{\frac{1}{3}} . P_2^{1-\frac{1}{3}} \right]$$

$$= u^{\frac{4}{3}} \left[\left(\frac{1}{2}\right)^{\frac{2}{3}} P_1^{\frac{1}{3}} . P_2^{\frac{2}{3}} + 2^{\frac{1}{3}} . P_1^{\frac{1}{3}} . P_2^{\frac{2}{3}} \right]$$

$$= u^{\frac{4}{3}} P_1^{\frac{1}{3}} . P_2^{\frac{2}{3}} \left[\left(\frac{1}{2}\right)^{\frac{2}{3}} + 2^{\frac{1}{3}} \right]$$

$$= u^{\frac{4}{3}} P_1^{\frac{1}{3}} . P_2^{\frac{2}{3}} (1.88988)$$

Now, find the cost of obtaining utility level (ψ_0, P^0) at prices P^0,

$$c(\psi_0, P_0) = 1.88988 . (15.811)^{\frac{4}{3}} . 4^{1/3} . 4^{\frac{2}{3}}$$

$$= 299.99$$

$$\simeq 300 .$$

$$c(\psi_1, p^1) = (18.803)^{\frac{4}{3}} \, 2^{\frac{1}{3}} \cdot 4^{\frac{2}{3}} \, (1.88988)$$

$$= 299.99$$

$$\simeq 300.$$

Similarly, for the other costs,

$$c(\psi_1, p^0) = (18.803)^{\frac{4}{3}} \, 4^{\frac{1}{3}} \cdot 4^{\frac{2}{3}} \, (1.88988)$$

$$= 377.98$$

$$c(\psi_0, p^1) = (15.811)^{\frac{4}{3}} \cdot 2^{\frac{1}{3}} \cdot 4^{\frac{2}{3}} \, (1.88988)$$

$$= 238.102$$

(iii) Equivalent Variation is given by

$$EV = c(\psi_1, p^0) - c(\psi_0, p^0)$$

$$= 377.98 - 300$$

$$= 77.98$$

Compensating variation is given by

$$CV = c(\psi_1, p^1) - c(\psi_0, p^1)$$

$$= 300 - 238.102$$

$$= 61.898$$

Consumer surplus, CS :

Given, $u(x_1, x_2) = x_1^{\frac{1}{3}} \cdot x_2^{\frac{1}{2}}$

$m^0 = 300$

Walrasian demand,

$$x_1 = \frac{m}{3p_1} = \frac{300}{3p_1} = \frac{100}{p_1}$$

$$CS = \int_2^4 \frac{100}{p_1} \, dp_1 = 100 \int_2^4 p_1^{-1} \, dp_1$$

$$= 100 \left[\ln p_1\right]_2^4$$

$$= 100 \left[\ln(4) - \ln(2)\right]$$

$$= 69.3147$$

We get above EV and CV using Expenditure function.

Since, EV > CS, CS > CV, the goods are normal good.

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Or, এইভাবেও করা যায়

CV and EV using compensated / Hicksian demand

Consider Hicksian demand for x_1

$$x_1 = u^{\frac{4}{3}} \left(\frac{1}{2}\right)^{2/3} \left(\frac{P_2}{P_1}\right)^{2/3}$$

Substituting utility level, u_1 and P^1 price, we get,

$$x_1 = (18.803)^{4/3} \cdot \left(\frac{1}{2}\right)^{\frac{2}{3}} \left(\frac{4}{P_1}\right)^{\frac{2}{3}}$$

$$= 79.37 \cdot P_1^{-\frac{2}{3}}$$

Now integrating this from 2 to 4 to obtain

$$\boxed{\int x^n dx = \frac{x^{n+1}}{n+1}}$$

$$EV = \int_2^4 79.37 \, P_1^{-\frac{2}{3}} \, dP_1$$

$$= 79.37 \int_2^4 P_1^{-\frac{2}{3}} \, dP_1$$

$$= 79.37 \left[\frac{P_1^{-\frac{2}{3}+1}}{-2/3+1}\right]_2^4$$

$$= 79.37 \left[3 P_1^{\frac{1}{3}}\right]_2^4$$

$$= 79.37 \times 3 \left(4^{\frac{1}{3}} - 2^{\frac{1}{3}}\right)$$

$$= 77.78$$

Substituting utility level, u_0 and P^2 price, we get,

$$x_1 = (15.811)^{\frac{4}{3}} \left(\frac{1}{2}\right)^{\frac{2}{3}} \left(\frac{4}{P_1}\right)^{\frac{2}{3}}$$

$$= (15.811)^{4/3} \left(\frac{1}{2}\right)^{2/3} \times 4^{2/3} \, P_1^{-2/3}$$

$$= 63 \, P_1^{-2/3}$$

$$CV = \int_2^4 63\, P_1^{-2/3} \cdot dP_1$$

$$= 63 \times 3 \left[P_1^{\frac{1}{3}} \right]_2^4$$

$$= 189 \left(4^{\frac{1}{3}} - 2^{\frac{1}{3}} \right)$$

$$= 61.89 .$$

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